

Rahul Kumar

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EDUCATION

Indian Institute of Information Technology Ranchi

2023 – 2027

CGPA: 7.04 / 10.00

Bachelor of Technology in Computer Science and Engineering

WORK EXPERIENCE

Back-End AI Engineering Intern, FlyRank AI (Remote)

July 2026 – Present

- Built scalable backend services using Node.js, Express.js, applying clean architecture and modular design principles.
- Developed secure RESTful APIs with CRUD operations, JWT authentication, route-level authorization, and middleware-driven validation.
- Integrated MongoDB with Mongoose ODM and containerized backend services using Docker for reproducible, one-command deployments.
- Built a compliant, production-style web scraper enforcing robots.txt and rate-limiting rules, extracting and cleaning HTML data into structured datasets.

AI Coding Expert – Project Dynamo, Handshake AI (Remote)

August 2026 – Present

- Designed production-inspired backend engineering tasks to evaluate frontier AI models on Node.js, Express.js, JWT authentication, REST APIs, and concurrent systems.
- Created debugging challenges involving refresh token rotation, race conditions (TOCTOU), token family revocation, and secure session management.
- Developed deterministic verification workflows using integration testing, concurrent request simulation, mocked clocks, and security-focused acceptance criteria.

PROJECTS

Quick.ai | AI-Powered Content Creation SaaS Platform 🔗 🔗

Tech Stack: React.js, Node.js, Express.js, PostgreSQL (NeonDB), Clerk Auth, Cloudinary, Gemini AI, Replicate AI, Tailwind CSS

- Built an AI-powered SaaS platform featuring **9 AI tools** (Article Writing, Image Generation, Background/Object Removal, Text-to-Video) integrating Gemini, Clipdrop, and Replicate APIs.
- Implemented **Clerk-based tiered access control** with free/premium usage limits, usage tracking, and feature gating for premium AI services.
- Designed a scalable **PostgreSQL schema** on NeonDB and integrated Cloudinary-powered media processing with a real-time community feed.

E-Commerce Platform | Full-Stack Online Shopping System 🔗 🔗

Tech Stack: React.js, Node.js, Express.js, MongoDB, JWT, Razorpay, Stripe, Cloudinary, Docker

- Developed a full-stack e-commerce platform with separate **User and Admin dashboards**, supporting product management, checkout, and order status tracking..
- Integrated **Razorpay, Stripe, and COD** payment workflows with secure verification and automatic order processing.
- Built secure RESTful APIs with **JWT authentication**, role-based authorization, Cloudinary image uploads, and MongoDB for persistent storage.

CoffeeBlend | Full-Stack Coffee Ordering & Table Booking Platform 🔗 🔗

Tech Stack: HTML, CSS, Vanilla JavaScript, Node.js, Express.js, MongoDB Atlas, JWT, PayPal SDK, Render, Vercel

- Built a full-stack coffee ordering and table booking platform using Node.js, Express.js, MongoDB Atlas, and JWT authentication with a modular REST API architecture.
- Integrated **PayPal Checkout** for secure payments and implemented authenticated user as well as guest table booking workflows.

TECHNICAL SKILLS

- **Languages:** C, C++, Java, Python, JavaScript, TypeScript, SQL, HTML, CSS
- **Frontend:** React.js, Next.js, Redux, Tailwind CSS, Bootstrap
- **Backend:** Node.js, Express.js, FastAPI, REST APIs, JWT Authentication, OAuth, NextAuth.js, Clerk Authentication, Mongoose ODM, WebSockets
- **Databases:** MongoDB, Mongoose, PostgreSQL, NeonDB (Serverless)
- **AI & APIs:** Prompt Engineering, LangChain, LangGraph, RAG, AI Agents, Gemini AI, Replicate AI, Clipdrop API
- **Libraries & Tools:** Cloudinary, Socket.IO, Axios, Multer, Nodemailer, bcrypt.js, pdf-parse
- **Cloud & DevOps:** Git, GitHub, Docker, Postman, Vercel, Render
- **Core CS:** DSA, OOP, DBMS, Operating Systems, Computer Networks

ACHIEVEMENTS AND CERTIFICATIONS

- Active open-source contributor at **GSSoC 2026** (GirlScript Summer of Code), contributing to real-world projects on GitHub.
- Completed **Deloitte Technology Job Simulation** on Forage, with hands-on tasks in software development and coding (July 2025).
- Certified in **DBMS** and **Operating Systems** by Scaler Topics (2025).